Adult Vaccines in Alberta: Pneumococcal, RSV, Influenza, Shingles and COVID-19

A pharmacist's evidence brief with patient counselling points. Baseline risk → risk after vaccination, how each vaccine is designed, how much it reduces hospitalization, how long protection lasts. Sources are hyperlinked at each mention (peer-reviewed journals and NACI/WHO/Alberta Immunization Policy documents where they are the primary source). Prepared 2 November 2026 for the 2026–27 season.

How to read the risk numbers

Baseline rates are population averages for the age band shown; an individual with COPD, heart failure, diabetes, immunosuppression or frailty sits well above them. "Risk after vaccination" applies the published relative reduction to the baseline rate; it is an illustration of magnitude, not an individual prediction. Hospitalization rates from surveillance under-count disease (RSV and pneumococcal pneumonia especially) because testing is incomplete.

1. Pneumococcal conjugate vaccine (Prevnar 20; Capvaxive)

1.1 Why it is needed

Streptococcus pneumoniae causes invasive pneumococcal disease (IPD: bacteremia, meningitis, bacteremic pneumonia) and, far more often, non-bacteremic pneumonia. Canadian IPD incidence rose to 10.2 per 100,000 in 2022 and 17.6 per 100,000 in adults ≥60; in adults ≥65 the leading serotypes in 2023 were 3 (14%), 22F (12%), 4 and 19A (Golden et al., CCDR 2026). Bacteremic pneumococcal pneumonia has a case-fatality rate of 5–7%, higher with age and comorbidity, and pneumococcal meningitis 22% in adults (Canadian Immunization Guide). All-cause pneumonia hospitalization in Canadians ≥65 runs at roughly 1,500–1,800 per 100,000 per year with an in-hospital case fatality of about 12% (Can Respir J 2016); pneumococcus is the most commonly identified bacterial cause.

1.2 How the vaccine is built ("conjugate" versus "polysaccharide")

The pneumococcal capsule is a polysaccharide, which on its own is a T-cell-independent antigen: it triggers short-lived antibody from B cells without germinal-centre help, no memory, weak responses in the very old, and hyporesponsiveness with repeat dosing. That is Pneumovax 23. A conjugate vaccine covalently links each capsular polysaccharide to a carrier protein (CRM197, a non-toxic diphtheria-toxin mutant), so polysaccharide-specific B cells can present carrier peptides to CD4 T cells, converting the response to T-dependent: affinity maturation, memory B cells, higher-avidity opsonophagocytic antibody and reduced nasopharyngeal carriage (Pollard et al., Nat Rev Immunol 2009). Prevnar 20 contains 20 serotypes (1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 12F, 14, 15B, 18C, 19A, 19F, 22F, 23F, 33F) at 2.2 µg each (6B 4.4 µg) on CRM197. Capvaxive (Pneu-C-21) is designed around adult serotypes: 3, 6A, 7F, 8, 9N, 10A, 11A, 12F, 15A, 15C, 16F, 17F, 19A, 20A, 22F, 23A, 23B, 24F, 31, 33F, 35B; it drops the paediatric serotypes that childhood programs have largely eliminated. In Canadian adults ≥65 (2019–2023) the serotypes shared by both vaccines caused 44–50% of IPD, Pneu-C-21-only serotypes 28–34%, and Pneu-C-20-only serotypes 8–14% (NACI statement, 2024).

1.3 How much it prevents disease

- **Randomized evidence (PCV13, the platform Prevnar 20 extends):** CAPiTA, 84,496 Dutch adults ≥65 (Bonten et al., NEJM 2015): efficacy 45.6% (95% CI 21.8–62.5) against vaccine-type pneumococcal community-acquired pneumonia, 45.0% against non-bacteremic vaccine-type pneumonia, and 75.0% (41.4–90.8) against vaccine-type IPD; no efficacy against all-cause pneumonia. Protection did not wane over the 5-year trial (Patterson et al., Trials Vaccinol 2016).
- **PCV20 and PCV21 were licensed on immunobridging** (opsonophagocytic titres non-inferior to PCV13/PPV23 for shared serotypes); NACI rates that evidence as moderate certainty (NACI 2024).

- **Real-world PCV20, 16.5 million US Medicare beneficiaries ≥65, 2022–24** (Lancet Infect Dis 2026): adjusted effectiveness 25.6% against all-serotype IPD, 23.3% against pneumococcal pneumonia, 15.2% against all-cause pneumonia, 7.4% against lower respiratory tract infection; effectiveness was highest at 65–74 (35% IPD) and lower at ≥85 (17%). A VA test-negative study of 34,009 pneumonia admissions found 20.4% (2.1–35.3) effectiveness against culture-confirmed pneumococcal pneumonia hospitalization in adults ≥65 (medRxiv preprint, Sept 2026; not yet peer-reviewed).

Outcome, adults ≥65	Baseline (unvaccinated)	Relative reduction	Risk after vaccination (illustrative)
Invasive pneumococcal disease, any serotype	≈18 per 100,000/yr (Canada, ≥60, 2022)	≈25% (PCV20, Medicare)	≈13 per 100,000/yr
Vaccine-type IPD	≈60–75% of the above	75% (CAPiTA PCV13)	≈3 per 100,000/yr for covered serotypes
All-cause pneumonia hospitalization	≈1,500–1,800 per 100,000/yr	≈15% (PCV20, Medicare)	≈1,300–1,500 per 100,000/yr
Vaccine-type pneumococcal pneumonia	—	45% (CAPiTA); 23% all pneumococcal pneumonia (Medicare)	—

Read across: the vaccine removes most risk from the serotypes it contains, but a large share of adult pneumonia is non-pneumococcal or from uncovered serotypes, which is why the all-cause numbers move only 15%.

1.4 Alberta program and schedule

- Publicly funded **Prevnar 20, one dose**: all adults ≥65 who have not had Pneu-C-20, and adults ≥18 with risk conditions, regardless of prior PPV23/PCV13/PCV15 (Alberta Blue Cross Benefact 1268). Capvaxive is available for private purchase and does not count as the funded dose. Transplant and CAR-T recipients are referred to Public Health.
- Spacing: ≥1 year after Pneumovax 23; ≥8 weeks after Prevnar 13/Vaxneuvance/Capvaxive (risk groups) or ≥1 year (healthy ≥65); 8 weeks acceptable if immunosuppression is imminent.
- NACI: one dose of Pneu-C-20 or Pneu-C-21 for all adults ≥65 and at-risk adults; re-immunization with the same conjugate vaccine is not recommended and most people should not receive both (NACI 2024).

Counselling points

- "This is a one-time pneumonia and bloodstream-infection vaccine. It's different from the old Pneumovax: it's a 'conjugate' vaccine that teaches your immune system to remember, so you don't need boosters."
- "It covers the 20 strains that cause most serious pneumococcal disease. It won't prevent every pneumonia, because many are caused by other germs, but it cuts invasive disease by about a quarter overall and by about three-quarters for the strains it contains."
- "If you had Pneumovax at 65, you still need this one, at least a year later."
- "Expect a sore arm for a day or two; serious reactions are rare. It can be given with your flu, COVID, RSV or shingles shots."

2. RSV vaccines (Abrysvo, Arexvy, mResvia)

2.1 Why it is needed

RSV is not a childhood-only virus. Canadian hospital data (2010–2023, excluding Québec) show average seasonal RSV-associated hospitalization of 10 per 100,000 at 60–69, 22 at 70–79 and 63 at ≥80, with in-hospital case fatality of 6–11% rising with age and ICU admission in 16% (Abrams et al., CCDR 2025). Because RSV is tested for far less often than influenza, these rates are lower bounds; NACI notes the burden may approach that of influenza in older adults. RSV also precipitates COPD and heart-failure exacerbations that are coded under those diagnoses.

2.2 How the vaccine is built (structure-based design)

RSV enters cells with its fusion (F) protein, which sits on the virion in a spring-loaded **prefusion** shape and snaps irreversibly into a **postfusion** shape during entry. For decades vaccines used postfusion-like F and failed. The breakthrough was solving the prefusion structure bound to a neutralizing antibody (McLellan et al., Science 2013a) and engineering disulfide and cavity-filling mutations (DS-Cav1) that lock F in the prefusion conformation (McLellan et al., Science 2013b). Prefusion-only epitopes (sites Ø and V) are the targets of most neutralizing antibody in human serum, so a stabilized prefusion antigen raises neutralizing titres many-fold more than postfusion F. All three licensed vaccines are built on this principle:

	Arexvy (GSK)	Abrysvo (Pfizer)	mResvia (Moderna)
Antigen	120 µg recombinant prefusion F (RSV-A)	60 µg RSV-A + 60 µg RSV-B prefusion F (bivalent)	50 µg mRNA encoding membrane-anchored prefusion F
Adjuvant	AS01E (MPL + QS-21 liposomes; half the Shingrix dose)	None	None (lipid nanoparticle)
Season-1 efficacy vs RSV lower-respiratory-tract disease	82.6% (96.95% CI 57.9–94.1) (Papi, NEJM 2023)	66.7% ≥2 symptoms; 85.7% ≥3 symptoms (Walsh, NEJM 2023)	83.7% ≥2 symptoms (Wilson, NEJM 2023)
Alberta funding	Private	Publicly funded product	Private

2.3 How much it prevents hospitalization

- **Real-world, test-negative, 26 US hospitals, 6,958 adults ≥60, two seasons** (Surie et al., JAMA 2025): vaccine effectiveness against RSV hospitalization 58% (95% CI 45–68) overall; 69% (52–81) when vaccinated in the same season and 48% (27–63) when vaccinated the previous season. Protection against ICU-level illness was also demonstrated.
- **Trials:** one dose of Arexvy gave cumulative efficacy of 62.9% (97.5% CI 46.7–74.8) against RSV lower-respiratory disease and 67.4% against severe disease over three seasons (Ison et al., Lancet Respir Med 2025); Abrysvo 77.8% against ≥3-symptom disease at the end of season 2 (Walsh, NEJM 2023).

Age band	Baseline RSV-associated hospitalization (Canada)	Risk after vaccination, season 1 (VE 69%)	Season 2 (VE 48%)
60–69	10 per 100,000/season	≈3 per 100,000	≈5 per 100,000
70–79	22 per 100,000/season	≈7 per 100,000	≈11 per 100,000
≥80	63 per 100,000/season	≈20 per 100,000	≈33 per 100,000
In-hospital death, ≥80	≈7 per 100,000/season	≈2 per 100,000	≈4 per 100,000

Absolute benefit is modest in a healthy 65-year-old and substantial in an 80-year-old with COPD or heart failure, which is why NACI and Alberta prioritize by age and risk rather than offering it universally at 60.

2.4 Alberta program and NACI position

- Funded (one lifetime dose, Abrysvo): community-dwelling adults ≥70; adults ≥60 in continuing care or supportive living; First Nations, Métis and Inuit adults ≥60; from 1 Oct 2026, adults 18–69 with lung or stem-cell transplant, chronic oxygen therapy or dialysis (Alberta.ca). Others may purchase any of the three products.
- NACI (April 2026): strong recommendation for all adults ≥75 and 65–74 at increased risk; any product; a single dose protects for at least three years and it is not yet known whether responses can be boosted (NACI summary; CCDR 2026). Rare Guillain-Barré signal ≥65 with the protein vaccines; benefit judged to outweigh.

Counselling points

- "RSV is the virus that sends babies to hospital, but after 70 it hospitalizes people as often as flu does, and about one in ten of those admitted don't go home."
- "One dose. It cuts the chance of an RSV hospital stay by roughly two-thirds in the first year and about half the next year, and trials show protection through at least three winters."
- "It is not a yearly shot. If you've had one RSV vaccine, you don't need another right now; that advice may change in a few years once booster studies finish."
- "Best time is early fall before RSV season, but it can be given any time of year. It can go with your flu and COVID shots."
- "Sore arm, tiredness and headache for a day or two are common. A very rare nerve condition (Guillain-Barré) has been seen in a few people per million; RSV infection itself also raises that risk."

3. Influenza vaccines (standard dose, Fluzone High-Dose, Fluad)

3.1 Why it is needed and what the baseline risk is

Before the pandemic influenza caused about 15,000 hospitalizations a year in Canada, more than any other respiratory virus (NACI 2026–27 statement, CCDR). Influenza-attributable respiratory hospitalization in adults ≥65 is estimated at 145 per 100,000 per year versus 26 per 100,000 at 45–64, and laboratory-confirmed hospitalization in 2022–23 was 136 per 100,000 in ≥65; over 80% of influenza deaths in H3N2 seasons occur in this age group (NACI supplemental guidance, adults ≥65).

3.2 How the vaccine is "modelled": the strain-selection pipeline

The WHO Global Influenza Surveillance and Response System (>120 national laboratories, six Collaborating Centres) sequences HA genes from viruses collected September–January, tests them in haemagglutination-inhibition and neutralization assays against post-infection ferret antisera (ferrets are immunologically naive, so each antiserum reflects a single virus), maps antigenic distance by cartography, tests sera from people vaccinated with the previous formulation, and runs fitness forecasts of which clades will dominate. WHO then names prototype viruses in February; manufacturers need about six months, which is why the vaccine is a February prediction about November (WHO full report, Feb 2026).

2026–27 composition (WHO, 27 Feb 2026; Alberta biological page): all three components changed. Egg-based (Fluzone, Fluviral, Fluad, Fluzone HD): A/Missouri/11/2025 (H1N1)pdm09, A/Darwin/1454/2025 (H3N2, subclade K, the variant that drove the early, heavy 2025–26 season), B/Tokyo/EIS13-175/2025 (B/Victoria, C.3.1). Cell-based (Flucelvax): A/Darwin/1415/2025 and B/Pennsylvania/14/2025 for the H3N2 and B components, because egg-grown isolates lose a glycosylation site that changes antigenicity (Zost et al., PNAS 2017). No B/Yamagata: not detected worldwide since March 2020.

3.3 Regular dose vs High-Dose vs Fluad

	Standard dose (Fluzone, Fluviral, Flucelvax)	Fluzone High-Dose	Fluad
Design	15 µg HA per strain	60 µg HA per strain (4×)	15 µg HA per strain + MF59 (squalene oil-in-water adjuvant)
Mechanism	Antigen alone	More antigen → higher titres	MF59 recruits innate cells, boosts antigen transport to the node and germinal-centre/Tfh activity → higher and broader titres (O'Hagan, Vaccine 2012)
Alberta 2026–27	Ages 6 months–64	Transplant candidates/recipients ≥18 (HSCT, SOT, CAR-T)	All adults ≥65
Evidence vs standard dose	—	rVE 24% vs lab-confirmed influenza, RCT n=31,989 (DiazGranados, NEJM 2014); rVE 32% vs lab-confirmed influenza hospitalization and 8.8% vs influenza/pneumonia hospitalization in 466,320 randomized adults (FLUNITY-HD, Lancet 2025)	No RCT vs standard dose on lab-confirmed outcomes; observational rVE 0–36% vs outpatient visits and consistent reductions in coded hospitalization (NACI)
Head-to-head	—	Equivalent: rVE adjuvanted vs HD 1.5% (−8.4 to 10.5) vs PCR-confirmed influenza and 9.1% (−4.0 to 20.4) vs influenza hospitalization/ED, 429,595 adults (JAMA Netw Open 2026)	Same study

	Standard dose (Fluzone, Fluviral, Flucelvax)	Fluzone High-Dose	Fluad
Reactogenicity	Least	More local pain	More local pain (grade ≥3 RR 3.4 vs SD)

NACI: high-dose, adjuvanted or recombinant vaccine should be offered over standard dose to adults ≥65, with no preference among them (NACI supplemental guidance). Alberta's 2025 move from Fluzone HD to Fluad for seniors is a procurement choice between equivalent-tier products.

3.4 How much it prevents hospitalization

- Pooled test-negative studies: vaccine effectiveness against laboratory-confirmed influenza hospitalization 37% (95% CI 30–44) in adults ≥65 and 51% at 18–64; for H3N2 in ≥65, 43% in matched seasons and 14% in drifted seasons (Rondy et al., J Infect 2017).
- Last season (drifted subclade K, vaccine mismatched) Canadian interim outpatient VE against H3N2 was still 40% (28–49), 25% in ≥65 (Separovic et al., Eurosurveillance 2026); Beijing 23.5% (Shen et al., Eurosurveillance 2026). This season's vaccine contains subclade K, so higher H3N2 effectiveness is expected; Canadian estimates arrive in late January.

Adults ≥65	Baseline	With standard-dose vaccine (VE ≈37%)	With Fluad/High-Dose (≈ additional 10–30% relative)
Influenza-attributable respiratory hospitalization	≈145 per 100,000/yr	≈90 per 100,000/yr	≈65–80 per 100,000/yr
Laboratory-confirmed influenza hospitalization (2022–23)	136 per 100,000	≈85 per 100,000	≈60–75 per 100,000

The enhanced-product column uses the 32% relative reduction in lab-confirmed influenza hospitalization from FLUNITY-HD at the upper end and the 9% all-influenza/pneumonia estimate at the lower end; absolute benefit is largest in the frail and those with cardiorespiratory disease.

3.5 Why protection fades

- Antibody peaks 2–4 weeks after vaccination then declines; effectiveness falls about 7% per month for A(H3N2) and B and 6–11% per month for A(H1N1) (Ferdinands et al., CID 2017); the odds of testing positive rise 16% per 28 days since vaccination (Ray et al., CID 2019).
- Three mechanisms: (1) waning serum antibody, fastest in older adults because of fewer long-lived plasma cells; (2) antigenic drift of the circulating virus away from the vaccine strain during the season; (3) inactivated vaccines generate little mucosal IgA or tissue-resident T-cell memory, so protection depends on circulating antibody that can decline.
- NACI's 2026–27 review: early-fall vaccination still covers most of the season; delaying is not warranted and coverage matters more than timing (CCDR 2026). Enhanced vaccines start from a higher titre, which partly offsets waning.

Counselling points

- "Flu puts about 1 in 700 people over 65 in hospital each year; the vaccine cuts that by roughly a third to a half, and the seniors' version (Fluad) does a bit better than the regular shot."
- "All three strains in this year's shot are new, including the H3N2 variant that caused last winter's bad season, so last year's vaccine isn't enough."
- "It takes two weeks to work and fades slowly over the winter, about 7–10% a month, so October–November is the right time; it still covers the January–February peak."
- "Fluad has an added ingredient that makes your immune system respond more strongly; expect a sorer arm than usual for a day. You cannot get flu from it; every injectable flu vaccine is inactivated."
- "Egg allergy is not a reason to skip it."

4. Shingles vaccine (Shingrix, recombinant zoster vaccine)

4.1 Why it is needed and the baseline risk

Herpes zoster is reactivation of latent varicella-zoster virus when VZV-specific cell-mediated immunity declines with age or immunosuppression. Lifetime risk is up to 30%; Canada sees roughly 130,000 cases, 17,000 cases of postherpetic neuralgia (PHN), 252,000 physician visits and 2,000 hospitalizations a year (NACI, CCDR 2026). In Ontario adults ≥65 the incidence is 59 per 10,000 per year, i.e. about 1 in 170 each year, rising with age (Buchan et al., PLoS One 2021); Canadian zoster hospitalization rates are 4–12 per 100,000 person-years (Letellier et al., CJIDMM 2018). PHN (pain persisting >90 days) follows 10–18% of cases and is the complication patients fear; ophthalmic zoster, stroke and disseminated disease are rarer.

4.2 How the vaccine is built

Shingrix is non-live. Each dose contains 50 µg of recombinant glycoprotein E (gE), the most abundant VZV envelope protein and the dominant target of VZV-specific CD4 T cells, combined with the AS01B adjuvant system (50 µg MPL, a TLR4 agonist, plus 50 µg QS-21 saponin in liposomes) (Alberta biological page). AS01 activates dendritic cells in the draining node and drives a strong Th1 CD4 response and polyfunctional memory T cells, which is exactly the arm of immunity that declines with age and permits reactivation (Didierlaurent et al., Expert Rev Vaccines 2017; Cunningham et al., J Infect Dis 2018). The live attenuated predecessor (Zostavax) boosted that immunity only modestly (≈51% efficacy, waning within 5 years) and could not be given to immunocompromised people; the recombinant design solves both problems.

4.3 How much it prevents shingles and its complications

- **ZOE-50** (15,411 adults ≥50): efficacy 97.2% (95% CI 93.7–99.0) against zoster (Lal et al., NEJM 2015). **ZOE-70** (pooled ≥70): 91.3% (86.8–94.5) against zoster and 88.8% (68.7–97.1) against PHN (Cunningham et al., NEJM 2016).
- **Complications:** 93.7% (59.5–99.9) against non-PHN complications (ophthalmic, disseminated, neurologic, vasculitis/stroke); 0 zoster-related hospitalizations in the vaccine arms versus 5 with placebo (Kovac et al., Vaccine 2018).
- **Real world, Medicare ≥65:** two-dose effectiveness 70.1% (68.6–71.5) against zoster, 68.5% at ≥80, 64.1% in the immunocompromised, and 76.0% (68.4–81.8) against PHN; one dose only 56.9% (Izurieta et al., CID 2021).
- **Immunocompromised adults (RCTs):** 68% in autologous HSCT, 87% in hematologic malignancy, 90.5% in immune-mediated disease (NACI 2026).

Adults ≥65	Baseline per year	Relative reduction	Risk after two doses (illustrative)
Shingles	≈590 per 100,000 (1 in 170)	91% (trial, ≥70); 70% (real world)	≈55 per 100,000 (trial) to ≈180 (real world)
Postherpetic neuralgia	≈60–100 per 100,000	89% (trial); 76% (real world)	≈10–25 per 100,000
Zoster hospitalization	4–12 per 100,000	≈94% vs complications; 0 vs 5 admissions in trials	<1–2 per 100,000

4.4 Alberta program and NACI position

- Alberta publicly funds Shingrix only for solid-organ transplant candidates/recipients, autologous HSCT recipients and CAR-T recipients ≥18 (Alberta biological page, Sept 2026); Veterans Affairs and NIHB cover some seniors (NIA summary). Everyone else buys it privately (two doses, 2–6 months apart; private drug plans often cover it).
- NACI recommends Shingrix for all adults ≥50 and for immunocompromised adults ≥18 (two doses 2–6 months apart; 1–2 months if immunosuppression is imminent, completed ≥14 days before it starts)

(NACI, CCDR 2026). Prior shingles or prior Zostavax are not reasons to withhold it; wait ≈1 year after an episode or after Zostavax.

- Reactogenicity is the main counselling issue: grade 3 (activity-limiting) reactions in 16.5% versus 3.1% with placebo, lasting 1–3 days (ACIP, MMWR 2018). Fever and shivering have been reported when co-administered with Pneumovax 23.

Counselling points

- "About 1 in 3 people get shingles in their lifetime, and after 65 the risk is about 1 in 170 every year. The nerve pain that can follow (PHN) lasts months and is hard to treat; that is the real reason to vaccinate."
- "Two doses, two to six months apart. Both are needed: one dose gives about 57% protection, two give over 90% in trials and about 70% in everyday use, and it has held for more than 10 years without a booster."
- "It is not a live vaccine, so it is safe if you are on immune-suppressing medication; in fact those patients are the ones Alberta funds it for."
- "Plan for a rough day or two: about 1 in 6 people get a sore arm, fatigue, muscle aches or chills strong enough to interfere with activities. That is the adjuvant doing its job. Book it when you don't have anything important the next day."
- "If you've had shingles or the old Zostavax, you should still get Shingrix, about a year later."

5. How long protection lasts: Prevnar, RSV, influenza, COVID-19, Shingrix

Vaccine	Durability evidence	Practical duration	Booster?
Prevnar 20 / Capvaxive	PCV13 efficacy stable over 5 years in CAPiTA with no waning (Patterson 2016); PCV20 Medicare follow-up 2 years (Lancet ID 2026); conjugate vaccines generate memory B cells (Pollard 2009)	Many years; treated as lifetime	No; re-immunization with the same conjugate not recommended (CIG)
RSV (all three)	Arexvy 82.6% → 48.0% by season 3, cumulative 62.9% (Ison 2025); Abrysvo 77.8% through season 2 (Walsh 2023); mResvia ≈50% at 18–24 months; real-world hospitalization VE 69% same season → 48% prior season (Surie 2025)	≥3 seasons, declining	Not yet; 12-month revaccination added nothing (Ison 2024); 24–60-month trials under way (IID 2026)
Influenza	≈7–10% relative decline per month (Ferdinands 2017; Ray 2019); strains change yearly	One season	Annual dose
COVID-19	Hospitalization/death VE ≈57% at 4 weeks → 34% at 20 weeks (JAMA Intern Med 2026, Nebraska); 51% at 7–59 days → not measurable at 6–10 months (Wiegand, JAMA Intern Med 2026); infection protection shorter	4–6 months for severe disease	Fall dose; second dose ≥3 months later strongly recommended for ≥80, LTC, immunocompromised, 65–79 with risk conditions (NACI fall 2026)
Shingrix	≥84.7% through year 4 (Cunningham 2016); 79.7% (73.7–84.6) over years 6–11 and 82% in year 11 (ZOE-LTFU final analysis; Strezova 2022)	≥11 years, still high	None recommended

Why they differ: pathogens that mutate fast (influenza, SARS-CoV-2) outrun antibody; those that don't (VZV, pneumococcal capsules) allow durable memory. Adjuvanted recombinant protein (Shingrix, Arexvy) and conjugate platforms build T-cell and memory-B-cell responses; inactivated influenza vaccine mostly builds short-lived antibody. RSV's prefusion-F vaccines are durable but the virus's antibody-evading entry route still lets protection drift down over seasons.

One-minute summary for the counter

	Pneumococcal	RSV	Influenza	COVID-19	Shingles
Who (Alberta funded)	≥65; ≥18 at risk	≥70; ≥60 in care/FNMI; 18–69 transplant/O2/dialysis	Everyone ≥6 mo; Fluad ≥65	≥65; risk groups; HCW; pregnancy	Transplant/CAR-T only (NACI: ≥50)
Doses	1, lifetime	1, lifetime (so far)	1 per year	1 per year (+1 for high-risk)	2, 2–6 months apart
Cuts hospitalization by	≈25% IPD; 15% all pneumonia	≈69% season 1; 48% season 2	≈37% (SD); more with Fluad/HD	≈50% early, ≈35% by 5 months	≈94% vs complications; 70–91% vs shingles
Lasts	Years	≥3 seasons	One season	4–6 months	≥11 years

Educational reference for pharmacist practice; verify eligibility and product supply against the current Alberta Immunization Policy biological pages and product monographs before administering.